Project Design Phase II
Solution Requirements (Functional & Non-functional)

Date	8 March 2026
Team ID	699d5fa0c40b5ad0e9ee5a7d
Project Name	Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis
Maximum Marks	4 Marks

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Input	User enters patient health parameters through the web form
FR-2	Data Processing	System validates and preprocesses the input data
FR-3	Hypertension Prediction	Machine learning model predicts blood pressure category
FR-4	Result Display	Prediction result is displayed on the web interface
FR-5	Model Integration	Flask backend loads the trained ML model for prediction
FR-6	Data Handling	System processes input data and sends it to the ML model

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should have a simple and user-friendly interface for entering patient data
NFR-2	Security	Input validation should be implemented to prevent invalid or malicious data
NFR-3	Reliability	The system should provide consistent prediction results without failure
NFR-4	Performance	Prediction results should be generated quickly after input submission
NFR-5	Availability	The system should be accessible through a web browser whenever required
NFR-6	Scalability	The system should be able to handle multiple prediction requests and future dataset expansion